
Global Convergence and Better Spectral Bias in Low-Rank Neural Networks

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Abstract

1 Neural networks often struggle to learn highly oscillatory functions at finite training
2 time: low-frequency components are fitted first, while high-frequency modes can
3 remain poorly recovered even when the model is expressive enough to represent
4 them. Low-rank networks are usually introduced as compressed alternatives to
5 dense models, but this view overlooks a more useful possibility: rank can act
6 as a structural control on what the network learns first. In this paper, we show
7 that this is the case. We first prove that low-rank random-feature networks in the
8 mean-field limit converge to a global minimizer of the population risk whenever
9 their limiting dynamics converge. We then show that the rank is not merely a
10 compression parameter: choosing it correctly can reduce the number of trainable
11 degrees of freedom while also improving the fit of highly oscillatory targets. The
12 key practical message is that the best rank is typically intermediate. If the rank is
13 too small, the model lacks expressivity; if it is too large, it recovers the finite-time
14 bias of the dense model. Controlled geometric and Fourier diagnostics, together
15 with high-frequency regression experiments, show that an appropriate low rank can
16 lower test loss, improve high-frequency recovery, and that the optimal rank shifts
17 with the target spectrum and training objective.

18 1 Introduction

19 Low-rank neural networks replace dense matrices by factored or bottlenecked operators. The standard
20 motivation is computational: fewer trainable parameters and cheaper memory traffic. This paper
21 argues that the more important question is mathematical. Can rank be used as a principled control on
22 training dynamics rather than only as a compression parameter?

23 This question is important for scientific machine learning. Many tasks in AI for science require
24 learning functions with oscillations, sharp transitions, multiscale structure, or high-frequency Fourier
25 content: wave fields, PDE solution operators, molecular potentials, turbulent signals, and inverse
26 problems all contain information at small spatial or temporal scales. Neural networks can represent
27 such functions, but finite-time training often learns smooth, low-frequency components first. This
28 frequency bias can be modified by the data distribution or by replacing the loss with a Sobolev
29 norm [23]; our question is whether the architecture itself, through rank, can provide another control
30 parameter. This makes the problem difficult: expressivity alone is not enough, and a model that is
31 globally trainable in principle can still fail to recover the high-frequency part under a realistic training
32 budget.

33 We answer the first question positively for a low-rank random-feature architecture. The model
34 freezes a rich random-feature map and trains low-rank channel weights through mean-field gradient
35 flow. Since the frozen features keep a dense span throughout training, the usual obstruction in deep
36 mean-field convergence proofs is removed: at a limit point, zero gradient implies the conditional

first-order condition for the loss, and the loss-identifiability assumption forces global optimality. Low rank enters through the mixing matrix and channel index, changing constants but not the logic of the proof.

The second question is more subtle. Global convergence is an asymptotic statement: it says what happens if the population dynamics settle. Spectral bias is a finite-time phenomenon: it asks which parts of the target are reached first under a limited optimization budget. These two views can disagree on oscillatory targets, because a model may be globally trainable and still spend most of training on the low-frequency part of the signal. We show that rank changes this finite-budget behavior by changing the geometry of ReLU networks, hence which Fourier modes are recovered early. The practical outcome is a rank-selection rule: choose the smallest rank that avoids the approximation bottleneck while improving high-frequency recovery.

Contributions. The paper makes three contributions:

- We give a globally convergent anchor model for low-rank networks. Freezing a rich random-feature map keeps the feature span from collapsing, while the trainable low-rank channels still allow a mean-field feature-learning interpretation.
- We explain why rank can change high-frequency learning. In ReLU networks, lowering rank reduces the number of independent path constraints, which changes the piecewise-affine geometry that controls Fourier energy.
- We provide controlled diagnostics for choosing rank. The experiments show that the useful rank is usually intermediate and moves with the target spectrum or training objective, so rank should be selected from spectral recovery rather than parameter count alone.

2 Related Work

This section positions the paper relative to mean-field convergence, low-rank adaptation, bottleneck-rank theory, and spectral bias.

Mean-field convergence. Mean-field theory analyzes neural network training through the deterministic evolution of parameter distributions in the large-width limit. Foundational work established global convergence mechanisms for two-layer models and Wasserstein gradient flows under suitable convexity, homogeneity, or support assumptions [17, 5]. Later multilayer mean-field analyses clarified how deep feature maps evolve and when stationary points can be related to global optima [20, 19]. Our result follows this line but focuses on low-rank random-feature networks: the existence and uniqueness estimates use the same mean-field template, while the convergence proof must be adapted to channel-wise low-rank couplings.

Low-rank and random features. Low-rank neural networks are often motivated by parameter efficiency, with LoRA-style decompositions now standard in large-scale adaptation [15]. Recent theory has begun to explain why LoRA can be trainable: Jang, Lee, and Ryu [16] show that, in the NTK fine-tuning regime, sufficiently large LoRA rank removes spurious local minima and yields low-rank solutions with good generalization. Most theory for low-rank networks focuses on expressivity, approximation, kernel behavior, or LoRA fine-tuning rather than global mean-field training dynamics. Random features provide a tractable bridge between finite networks and kernel methods: fixing the feature map turns the representation into a stable basis, while training the channel weights retains a mean-field optimization problem. In our RF-LR setting, frozen random features prevent support collapse and preserve the dense-span property needed for the stationarity-to-optimality argument; the low-rank factors enter through bounded channel mixing rather than through a full dense contraction chain.

Bottleneck rank and implicit low-rank bias. Recent work gives a functional notion of optimal rank that is closer to our setting than the algebraic rank of a single matrix. Jacot [12] studies the implicit bias of large-depth homogeneous networks and shows that their representation cost converges toward a nonlinear rank notion, sandwiched between Jacobian rank and bottleneck rank. The bottleneck rank is the smallest inner dimension k for which a function can be factored as $f = h \circ g$ through $\mathbb{R}^k$. Follow-up work [13] proves that learned deep representations exhibit a bottleneck structure

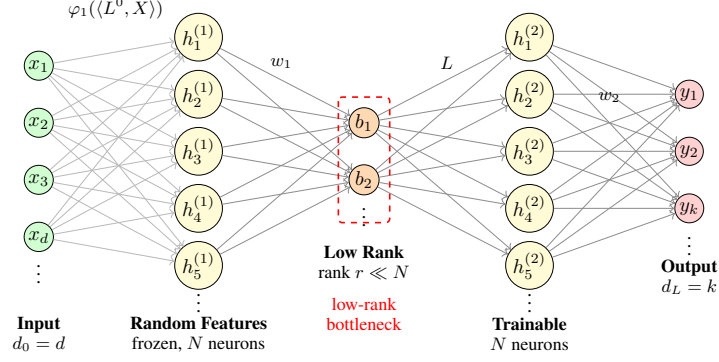


Figure 1: Architecture of a three-layer low-rank random-feature network with frozen random features, a rank- r bottleneck, and trainable output weights.

and introduces finite-depth corrections that balance low inner dimension against regularity. This suggests a theoretical target rank: the smallest rank that captures the intrinsic bottleneck dimension of the target without forcing an irregular or rank-underestimating interpolant. Complementarily, Bantzis, Simon, and Jacot [3] show that the first saddle escape of deep ReLU networks has a low-rank bias in deeper layers and propose saddle-to-saddle dynamics with increasing bottleneck rank. Our rank-selection principle is consistent with this picture: training should start from low effective rank, but useful learning requires increasing rank until approximation and spectral recovery saturate.

Spectral bias. The spectral-bias literature identifies why neural networks often fit low-frequency components before high-frequency components. Rahaman et al. [21] connect this learning order to Fourier properties of piecewise linear networks, while Yu, Yang, and Townsend [23] show that the bias can be tuned through nonuniform data and Sobolev-type losses. This paper keeps that Fourier viewpoint but changes the intervention: instead of reweighting data or losses, we study rank as an architectural parameter that changes the geometry seen by finite-time training.

3 Freezing Half the Weights Allows Global Convergence

This section defines the RF-LR mean-field model and states the conditional global convergence theorem.

Let $X \in \mathbb{R}^d$ and Y be drawn from a population distribution. The finite-width low-rank random-feature network used throughout the paper is the one from the original experiments:

$$h^{(0)}(x) = x, \quad h^{(\ell)}(x) = \frac{1}{n_\ell} \sum_{j=1}^{n_\ell} w_j^{(\ell)} \varphi_\ell \left(\langle L_j^{(\ell)}, h^{(\ell-1)}(x) \rangle + b_j^{(\ell)} \right), \quad (1)$$

where the feature directions $L_j^{(\ell)}$ and biases $b_j^{(\ell)}$ are frozen, while the channel weights $w_j^{(\ell)}$ are trained. This is a low-rank factorization of a dense layer in which the left factor is fixed: if $W_\ell = U_\ell V_\ell^\top$ with $U_\ell \in \mathbb{R}^{n_\ell \times r_\ell}$ bounded and full column rank, then U_ℓ plays the role of a frozen random-feature or mixing basis and the trainable factor V_ℓ is absorbed into the channel weights. Thus one may avoid training U_ℓ entirely; drawing it as a bounded full-rank tall-skinny matrix is enough for the theory, while choosing $U_\ell \in \text{St}(n_\ell, r_\ell)$ is an optional numerical normalization.

3.1 Mean-Field Formulation

The mean-field viewpoint is the relevant one for global convergence. Pure kernel or neural-tangent analyses give convexity by freezing the representation, but they largely remove feature learning. Finite-width nonconvex analyses allow feature learning but rarely provide global convergence guarantees for deep low-rank networks. Mean-field theory is currently the most developed framework that can keep a feature-learning interpretation while still proving that limiting gradient-flow trajectories reach global minimizers under identifiability assumptions. We use a frozen-feature low-rank specialization

because it is the cleanest setting in which the dense-span argument is preserved; it should be viewed as the globally convergent anchor case for more general low-rank feature-learning models.

In the mean-field limit, frozen feature maps are written $L^0(C_\ell)$ and trainable channel weights are written $w_\ell(t, C_\ell)$. The channel mixer $L \in \mathbb{R}^{N_2 \times r}$ is a bounded full-column-rank tall-skinny factor, not necessarily an orthogonal matrix. We write the second-layer operator as $W_2 = LA_2^\top$, with L fixed and A_2 trainable, then absorb the small right factor $A_2^\top$ into the rank-channel functions below. One may additionally choose $L \in \text{St}(N_2, r)$, or periodically re-orthogonalize it, as a numerical design choice; this is natural for matrix-structured optimizers such as Muon, which orthogonalize matrix updates by Newton–Schulz or polar iterations [14, 4]. In the two-hidden-layer case, the first partial functions are

$$f_k(t, x) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)], \quad k = 1, \dots, r, \quad (2)$$

and the second-layer preactivation is

$$H_2(t, c_2; x) = \sum_{k=1}^r L_{c_2, k} f_k(t, x). \quad (3)$$

Thus H_2 is the reconstruction of the second-layer signal from a bounded rank- r channel basis. The output is obtained by averaging the trainable top weights against $\varphi_2(H_2)$. For deeper networks, the same alternating pattern is used: trainable channel weights are averaged against frozen random features and mixed through bounded low-rank channel matrices.

The population objective is

$$\mathcal{L}(W) = \mathbb{E}_{(X, Y)} [\mathcal{L}(Y, \hat{y}(X; W))]. \quad (4)$$

The mean-field gradient flow evolves the trainable weights by

$$\partial_t w_\ell(t, \cdot) = -\xi_\ell(t) \nabla_{w_\ell} \mathcal{L}(W(t)), \quad (5)$$

where ξ_ℓ are bounded learning-rate schedules. The exact formulas are standard backpropagation in the mean-field variables; the only low-rank modification is the channel mixing through $L_{c_\ell, k}$.

Under the technical regularity, diversity, loss-identifiability, and convergence assumptions stated in Appendix A, freezing the feature directions gives a well-defined mean-field system and allows the usual stationarity argument to conclude global optimality.

Theorem 3.1 (Well-posedness and conditional global convergence of RF-LR training). *Under Assumptions A.1–A.6, the low-rank mean-field gradient-flow system has a unique solution on every finite interval $[0, T]$ and therefore on $[0, \infty)$. Moreover, if the resulting low-rank mean-field dynamics converges to W^* in the modified $\mathcal{W}_4$ coupling topology defined in Appendix A through Assumption A.6, then W^* is a global minimizer of the population loss:*

$$\mathcal{L}(W^*) = \inf_W \mathcal{L}(W). \quad (6)$$

For every depth $L \geq 2$, this holds with standard independent initialization of the trainable weights.

The convergence statement is conditional: as in the corresponding multilayer mean-field results, we do not prove that every trajectory converges. The theorem says that convergence cannot occur to a bad stationary point under the stated dense-span, non-degeneracy, and loss-identifiability assumptions. The well-posedness part only justifies that the limiting dynamics are genuine ODEs rather than formal flows. This is the natural Picard argument of Nguyen and Pham [19]: nothing structural changes in the proof, except that dense-layer operator norms are replaced by bounded low-rank mixing constants. The complete well-posedness proof is deferred to the supplementary material, Appendix B.

Proof idea. The existence and uniqueness part follows the standard mean-field fixed-point argument, whose full details are given in Appendix B. We only summarize the convergence-to-global-minimum step here; Appendix C gives the full notation and the five formal steps. We assume that gradient descent, or its mean-field gradient-flow limit, has converged somewhere: $W(t) \rightarrow \bar{W}$ in the modified $\mathcal{W}_4$ coupling topology defined in Appendix A. Stationarity and the persistent dense span of the frozen first-layer features then imply the integrated identity

$$\mathbb{E} \left[d_L(Z; \bar{W}) B_k^{(2)}(X; \bar{W}) \mid X = x \right] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x, \quad k = 1, \dots, r, \quad (7)$$

where $d_L(Z; \bar{W}) = \partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W}))$ and $B_k^{(2)}$ is the channel-wise backpropagated factor defined in Appendix C.

Step 3: remove the backpropagated factor. The non-degeneracy assumptions exclude a dead-channel limit: for $\mathbb{P}_X$ -a.e. x , at least one channel has a nonzero backpropagated factor $B_k^{(2)}(x; \bar{W})$. The identity we use keeps this factor inside the conditional expectation:

$$\mathbb{E}[d_L(Z; \bar{W}) B_k^{(2)}(X; \bar{W}) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x, \quad k = 1, \dots, r.$$

Since $B_k^{(2)}(X; \bar{W})$ is X -measurable, the non-degeneracy assumption then allows us to choose at least one nonzero backpropagated factor at almost every x , so

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W})) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x. \quad (8)$$

Step 5: the trajectory loss converges to the limit loss. Since $W(t) \rightarrow \bar{W}$ in the modified $\mathcal{W}_4$ coupling topology defined in Appendix A, there are couplings π_t between $W(t)$ and $\bar{W}$ whose weighted channel gaps vanish. In the two-layer channel notation, the key convergence statement is exactly

$$\mathbb{E}_{\pi_t} \left[(1 + |\bar{w}_2|) |\bar{w}_2| \sum_{k=1}^r |\bar{w}_{1,k}| |w_{1,k} - \bar{w}_{1,k}| \right] \rightarrow 0, \quad (9)$$

with analogous quantities for all deeper layers. The forward stability estimate uses the ReLU channel expansion of a low-rank preactivation,

$$\left| \text{ReLU} \left(\sum_{k=1}^r x_k \right) - \text{ReLU} \left(\sum_{k=1}^r y_k \right) \right| \leq \sum_{k=1}^r |x_k - y_k|. \quad (10)$$

Together with the low-rank form $H_2 = \sum_k L_{c_2,k} f_k$, (10) bounds every forward and backward difference by the same channel integrals. Hence

$$\mathbb{E}[|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \rightarrow 0. \quad (11)$$

Lipschitzness of the loss in its second argument on the a priori bounded trajectory yields

$$|\mathcal{L}(W(t)) - \mathcal{L}(\bar{W})| \leq K \mathbb{E}[|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \rightarrow 0. \quad (12)$$

Step 4: identify the limit as globally optimal. By the loss-identifiability condition in Assumption A.3, namely (26) in Appendix A,

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, u) \mid X = x] = 0 \implies \mathbb{E}[\mathcal{L}(Y, u) \mid X = x] = 0. \quad (13)$$

Applying this with $u = \hat{y}(x; \bar{W})$ and using non-negativity of $\mathcal{L}$ gives

$$\mathcal{L}(\bar{W}) = 0 = \inf_W \mathcal{L}(W). \quad (14)$$

Together with Step 5, $\mathcal{L}(W(t)) \rightarrow \mathcal{L}(\bar{W})$. $\square$

4 Low-Rank Spectral Bias

This section explains how low-rank path constraints change CPWL geometry and the Fourier coefficients that control finite-resolution spectral bias.

Connection to spectral bias. Theorem 3.1 says that a convergent trajectory reaches a global minimizer, but it does not describe the order in which Fourier modes are recovered at finite time. This section explains the architectural mechanism behind that order: low rank constrains active path coefficients and switching geometry, so it changes the CPWL face moments that control finite-resolution Fourier transfer. Thus the role of Rahaman et al. [21] here is not only as prior related work, but as the Fourier template that our rank-dependent face-moment calculation refines.

In a ReLU network, on every activation region the network is affine, and the affine coefficient is a sum over active paths. We use the same path-product viewpoint as in path-based analyses of loss

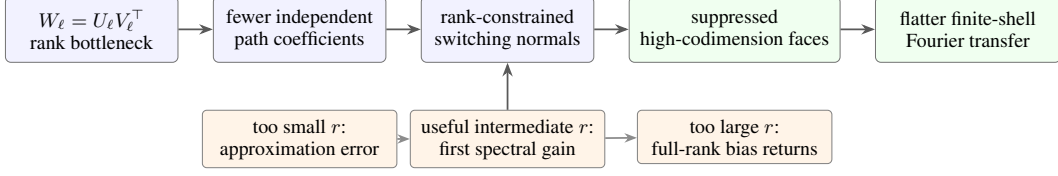


Figure 2: Mechanism summary linking low-rank factorization, path coefficients, switching normals, CPWL faces, and finite-shell Fourier transfer.

surfaces and finite-width neural kernels [6, 10]. For a scalar-output depth- L network, this can be written schematically as

$$f(x) = \sum_{p=(i_0, \dots, i_L)} \left(\prod_{\ell=1}^L W_{i_\ell i_{\ell-1}}^{(\ell)} \right) \left(\prod_{\ell=1}^{L-1} \mathbf{1}_{\{h_{i_\ell}^{(\ell)}(x) > 0\}} \right) x_{i_0}, \quad (15)$$

where p ranges over input-to-output paths. If $W_\ell = U_\ell V_\ell^\top$ has rank r_ℓ , then many symbolic paths share the same latent channels, so their coefficients cannot vary independently. These path constraints also constrain ReLU switching normals $\nabla_x h_j^{(\ell)}(x)$: the normals lie in a smaller span than in a dense layer. Geometrically, this suppresses low-dimensional, high-codimension intersections; spectrally, the Fourier shell law below predicts more visible facet-dominated high-frequency mass.

Mechanism summary. Figure 2 summarizes the mechanism. A low-rank factorization reduces independent path coefficients in (15); this constrains switching normals, suppresses high-codimension CPWL faces, and moves shell-averaged Fourier energy toward the facet-dominated regime of Rahaman et al. [21]. The useful rank is intermediate: too small gives approximation error, while too large recovers the full-rank low-frequency-first bias. A frequency shell means modes with comparable radius $\|\xi\| \simeq \rho$.

This connects to bottleneck rank: the population lower bound is the smallest inner dimension through which the target can factor [12, 13], while the finite-budget choice also balances optimization and spectral recovery. In experiments, we estimate this by the first rank where approximation is controlled and Fourier recovery no longer improves.

This algebraic constraint becomes geometric. Inside a cell of the previous layers, a new switching normal has the form

$$n_{\ell i \varepsilon} = B_{\ell-1, \varepsilon}^\top V_\ell u_{\ell i} \in \text{Im}(B_{\ell-1, \varepsilon}^\top V_\ell), \quad \dim \text{Im}(B_{\ell-1, \varepsilon}^\top V_\ell) \leq \min(d, r_1, \dots, r_\ell). \quad (16)$$

Thus rank lowers the dimension of the normal subspace and suppresses high-codimension faces first.

Face Fourier expansion. Let $g = \chi f$, where f is CPWL on a finite polyhedral complex in $\mathbb{R}^d$ and $\chi \in C_c^\infty(\mathbb{R}^d)$ is an observation window. Let $\mathcal{F}_q(f)$ be the codimension- q faces of the complex. For every large frequency $\xi = \rho\theta$, $\theta \in \mathbb{S}^{d-1}$, the Fourier transform has the expansion

$$\widehat{g}(\rho\theta) = \sum_{q=1}^d \rho^{-(q+1)} \sum_{F \in \mathcal{F}_q(f)} A_F(\rho, \theta) + O(\rho^{-(d+2)}), \quad (17)$$

where A_F is a bounded oscillatory face coefficient controlled by the jump Δ_F of the appropriate normal derivative across the local fan around F . In particular,

$$|A_F(\rho, \theta)| \leq C_\chi \|\Delta_F\| \text{vol}_{d-q}(F) \omega_F(\theta). \quad (18)$$

This is the anisotropic phenomenon isolated by Rahaman et al. [21]: in almost all directions a ReLU network has fast $k^{-(d+1)}$ amplitude decay, while in special directions orthogonal to facets of the linear regions the decay can be as slow as k^{-2} . Thus low codimension faces are precisely the geometric structures that keep high-frequency Fourier mass visible at finite resolution.

Proposition 4.1 (Fourier shell law for CPWL networks). *Let $g = \chi f$ be a windowed continuous piecewise affine network in dimension d . Define the codimension- q face moment*

$$M_q(f) = \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \text{vol}_{d-q}(F)^2 \mathbb{E}_{\theta \in \mathbb{S}^{d-1}} \omega_F(\theta)^2. \quad (19)$$

220 Then the shell-averaged Fourier energy satisfies

$$S_f(\rho) = \mathbb{E}_{\|\xi\| \simeq \rho} |\widehat{g}(\xi)|^2 \lesssim \sum_{q=1}^d M_q(f) \rho^{-2(q+1)}. \quad (20)$$

221 Low rank suppresses M_q for larger q first, moving finite-resolution spectra toward the facet-dominated
 222 ρ^{-4} energy regime. This should be read as a coefficient statement rather than a claim that low rank
 223 creates a new universal exponent: the useful comparison with full rank is the reduction of M_q/M_1
 224 and of the transition coefficient $(M_q/M_1)\rho^{-2q+2}$ for $q \geq 2$.

225 *Proof idea.* Write f on its polyhedral complex as an affine function on each cell and set $g = \chi f$ with
 226 a smooth compactly supported window. The Fourier transform is then a sum of ordinary oscillatory
 227 integrals over cells. Integrating by parts in the direction $\theta = \xi/\|\xi\|$ produces boundary terms on cell
 228 facets. The order- ρ^{-1} traces cancel across interior facets because the CPWL network is continuous
 229 and the same window multiplies both sides.

230 The first non-canceling term is therefore the jump of the directional derivative across a facet, hence it
 231 carries the factor ρ^{-2} . Repeating the same integration-by-parts argument on the induced face complex
 232 localizes the next non-smooth terms on codimension- q faces, with amplitude order $\rho^{-(q+1)}$. This
 233 gives the face expansion (17), with coefficients controlled by the corresponding normal-derivative
 234 jumps and face volumes. This is the same anisotropic mechanism emphasized by Rahaman et
 235 al. [21]: generic directions decay quickly, while directions close to facet normals can retain slow
 236 $|\xi|^{-2}$ amplitude decay.

237 Squaring the expansion and averaging over a frequency shell gives diagonal face energies of order
 238 $M_q(f)\rho^{-2(q+1)}$. Oscillatory cross terms are either lower order by angular non-stationary phase or
 239 absorbed into the same moments by Cauchy–Schwarz. Thus the exponent is determined by codimen-
 240 sion, while rank changes the coefficients $M_q(f)$ by suppressing high-codimension intersections. A
 241 detailed integration-by-parts proof is given in Appendix D. $\square$

242 5 Experiments

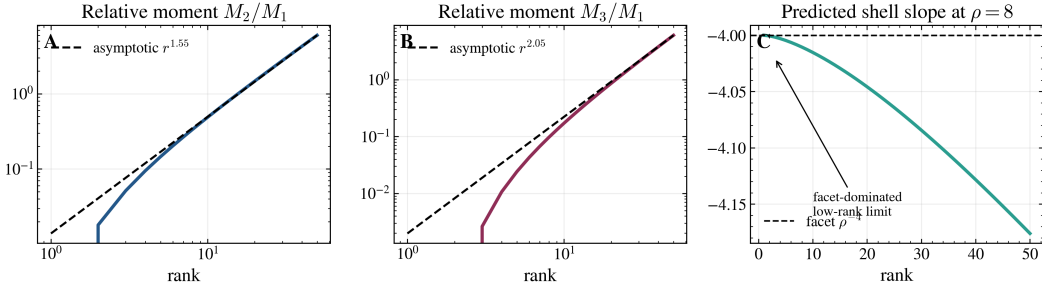


Figure 3: Theory proxy for the Fourier shell law. Panels A–B show relative face-moment proxies M_2/M_1 and M_3/M_1 , i.e. how much higher-codimension geometry is present relative to facets. Panel C shows the induced finite-shell slope, separating the universal codimension exponents from the rank-dependent coefficients.

243 This section presents controlled diagnostics that test the proposed geometric and Fourier mechanism,
 244 together with supporting high-frequency regression evidence.

245 We keep the experimental evidence deliberately short and defer the full setup of every generated plot
 246 to Appendix E. The main figures are controlled CPWL and Fourier/kernel-limit diagnostics: they test
 247 the mechanism predicted by Proposition 4.1, namely that rank changes the face-moment coefficients
 248 $M_q(f)$ and the finite-time recovery of Fourier modes. Separate finite-network MMNN sweeps on
 249 high-frequency targets are used only as supporting evidence that low-rank models can fit oscillatory
 250 functions and that optimizer choice and feature freezing matter in practice.

251 The experiments should therefore be read as mechanism checks rather than as a benchmark claim
 252 that one optimizer or one rank universally dominates. The CPWL plots isolate geometry, the

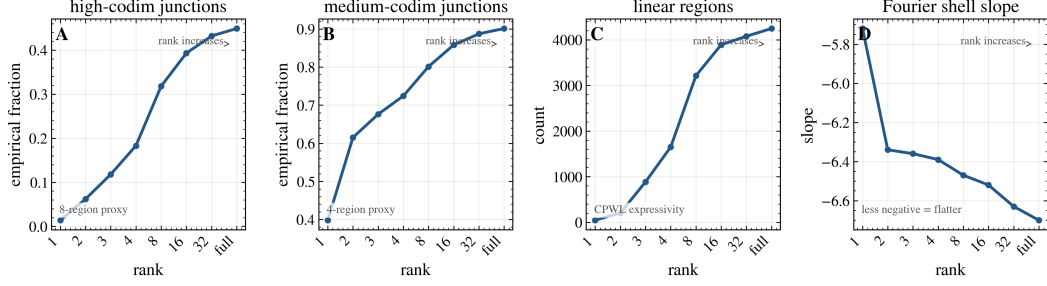


Figure 4: Three-dimensional CPWL geometry proxy. Panels A–B show the fraction of sampled neighborhoods where many affine regions meet, using thresholds of 8 and 4 regions as empirical high- and medium-codimension junction proxies. Panel C shows the number of linear regions, and Panel D shows the estimated shell slope.

253 Fourier/kernel plots isolate spectral transfer at fixed budget, and the rank-shift control tests whether
 254 the observed optimum is task-dependent. This separation is intentional: the theorem is an asymptotic
 255 convergence statement, whereas the spectral-bias claims concern finite-resolution coefficients and
 256 finite-time recovery.

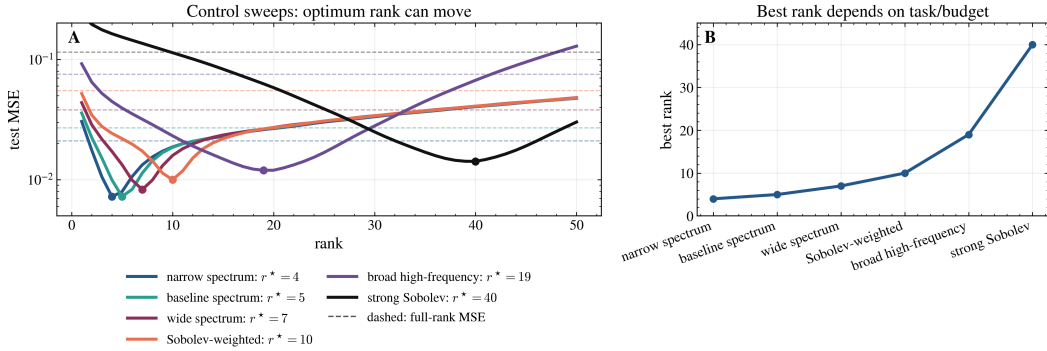


Figure 5: Rank-shift control. The left panel shows finite-budget MSE versus rank for several target/weighting regimes; the right panel shows the best rank in each regime. Dashed lines are full-rank baselines. The plot emphasizes that the useful rank moves with the spectrum and objective rather than staying fixed.

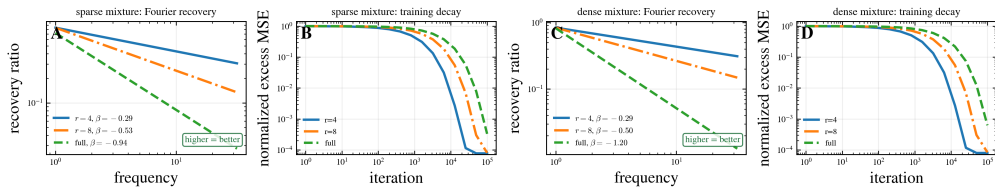


Figure 6: Mode-wise recovery and training curves. Panels A and C show Fourier recovery ratios $R_r(k, t)$ across frequency modes: higher and flatter curves mean that high frequencies are recovered more evenly relative to low frequencies. Panels B and D show normalized excess MSE during training, so the useful ranks are those that combine low error with flatter spectral recovery.

257 The one-dimensional recovery plots require a small caveat. A 1D ReLU network is a linear spline: if
 258 Δ_j denotes the derivative jump at knot t_j , then

$$\hat{f}(k) = -\frac{1}{(ik)^2} \sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \quad |\hat{f}(k)|^2 \sim |k|^{-4} A(k). \quad (21)$$

259 Thus the raw asymptotic tail stays near the classical k^{-4} law, and the finite-budget diagnostic should
 260 be mode-wise recovery,

$$R_r(k, t) = \frac{|\hat{f}_{r,t}(k)|}{|\hat{f}^*(k)|}, \quad \beta_r(t) = \frac{d \log R_r(k, t)}{d \log k}. \quad (22)$$

261 A flatter fitted slope $\beta_r(t)$ means that high modes are recovered more evenly relative to low modes
 262 on the plotted frequency window. Figure 6 reports this recovery ratio together with training MSE.

263 The geometry plots should be read through the face-moment shell law

$$S_f(\rho) \lesssim M_1(f)\rho^{-4} + M_2(f)\rho^{-6} + M_3(f)\rho^{-8} + \dots. \quad (23)$$

264 Figure 3 tests the coefficient prediction directly: higher-codimension moments grow with rank,
 265 so the finite-shell slope moves away from the facet-dominated regime. Figure 4 checks the same
 266 mechanism geometrically: high-codimension junctions are empirical proxies for the higher moments
 267 $M_2, M_3, \dots$, and they become more frequent as rank creates more independent switching geometry.
 268 Appendix E gives the full interpretation.

269 The spectral-bias experiment isolates the rank effect in a width- 2^{15} Fourier/kernel surrogate. Figure 5
 270 is the key control: the best finite-budget rank is intermediate, but it is not fixed at one value. It moves
 271 from small ranks on narrow or baseline spectra to larger ranks under wider spectra and Sobolev-
 272 weighted objectives used to tune frequency bias [23]. Across these regimes, the best low-rank point
 273 also beats the corresponding full-rank dashed baseline. Figure 6 shows why MSE alone is not enough:
 274 the useful rank also keeps high-frequency recovery flatter than the full endpoint.

275 Together, Figures 3–6 support the same conclusion: rank is not only a compression parameter. It
 276 changes the finite-budget spectral transfer, and the best rank is the first one that controls approximation
 277 while preserving flatter high-frequency recovery.

278 6 Conclusion

279 Low rank is not only a compression device. In the random-feature mean-field regime, it preserves
 280 global convergence because frozen features maintain dense span and low-rank channel mixing only
 281 changes constants. In ReLU and CPWL regimes, it also changes spectral bias through path-space
 282 constraints and Fourier face moments. This matters for AI for science because oscillatory targets are
 283 common in PDEs, waves, inverse problems, and molecular or multiscale modeling. Prior work shows
 284 that frequency bias can be tuned by data or Sobolev losses [23]; our contribution is to show that
 285 architecture provides another lever. The message is conservative but actionable: use the frozen-feature
 286 low-rank model as a globally convergent anchor, then choose rank by finite-budget spectral recovery
 287 rather than by parameter count alone.

288 The mechanism is the following. A globally convergent dynamic can still learn low frequencies first
 289 and fail at finite time on highly oscillatory targets. Low rank changes this finite-time behavior because,
 290 in ReLU networks, it constrains active path coefficient tensors; these constraints restrict switching
 291 normals, suppress high-codimension faces in the continuous piecewise affine complex, and change
 292 Fourier shell behavior. Here CPWL means continuous piecewise affine: the input space is split into
 293 polyhedral regions, and the network is affine on each region. In one dimension, the raw spline tail
 294 remains close to the classical k^{-4} law, so the right diagnostic is not the asymptotic exponent alone but
 295 the learned Fourier recovery curve. This leads to the practical rule used throughout the experiments:
 296 choose the smallest rank that avoids the approximation bottleneck while flattening the finite-time
 297 spectral transfer.

298 Several directions remain open. First, the convergence result should be extended from frozen
 299 random features to trainable low-rank factors, where the dense-span argument can fail if the features
 300 collapse. Second, the CPWL shell law suggests a measurable rank-dependent coefficient theory,
 301 but sharper finite-width estimates are needed to predict the best rank without a sweep. Third, the
 302 experiments indicate that optimizer geometry matters, especially for Muon-type orthogonalized
 303 updates; understanding how such updates interact with low-rank spectral transfer is a natural next
 304 step.

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A Assumptions and Dense-Span Property

This section records the assumptions used by the convergence theorem and fixes the notation for the mean-field limit.

We collect the assumptions used in Theorem 3.1. We reuse the multilayer mean-field framework and notation of Nguyen and Pham [19]: the mean-field width limit sends $n_1, \dots, n_{L-1} \rightarrow \infty$ at fixed depth L , and discrete neuron indices are replaced by labels $C_\ell \in \Omega_\ell$ with laws ρ_ℓ . Our only architectural change is that dense contractions are replaced by frozen random features and bounded low-rank channel mixers. The notation $\partial_2 \mathcal{L}(y, u)$ denotes the derivative of the loss with respect to its prediction argument.

Assumption A.1 (Bounded activations and low-rank mixing). *There exists $K \geq 1$ such that the activations φ_ℓ are K -Lipschitz, the activation derivatives used in backpropagation are bounded on the active region, and the low-rank mixers satisfy*

$$\sup_{c_\ell} \sum_{k=1}^{r_\ell} |L_{c_\ell, k}^{(\ell)}| \leq r_\ell K. \quad (24)$$

The mixers are full-column-rank tall-skinny factors in finite width. Orthogonality, $L^{(\ell)\top} L^{(\ell)} = I_{r_\ell}$, is an optional Stiefel normalization rather than a theorem assumption. The preactivations remain in the a priori bounded regime on every finite interval, and the learning-rate schedules are non-negative, bounded, and locally integrable.

Assumption A.2 (Sub-Gaussian initialization). *The trainable initial weights have sub-Gaussian tails, uniformly over channels. In the two-hidden-layer notation,*

$$\sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w_1^0(C_1, k)|^m]^{1/m} \leq K, \quad \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_2} [|w_2^0(C_2)|^m]^{1/m} \leq K. \quad (25)$$

The L -layer version imposes the same moment growth bound on every trainable channel. This assumption includes the standard initializations used in practice. In particular, Xavier or Glorot uniform initialization [9], with entries sampled independently from

$$\text{Unif} \left[-\sqrt{\frac{6}{d_{\text{in}} + d_{\text{out}}}}, \sqrt{\frac{6}{d_{\text{in}} + d_{\text{out}}}} \right],$$

is bounded and therefore sub-Gaussian after the usual fan-in/fan-out scaling. Xavier normal initialization, and more generally any independent centered sub-Gaussian initialization with the same variance scale, also satisfies the displayed moment bound. Thus the phrase “standard independent initialization” in Theorem 3.1 covers Glorot/Xavier uniform, Glorot/Xavier normal, and the usual sub-Gaussian or subnormal variants used for trainable weights.

Assumption A.3 (Data distribution and loss). *Inputs are bounded, $|X| \leq K$ almost surely, and the frozen first-layer features satisfy $\|L^0(c_1)\| \leq K$. The loss is non-negative, and $\partial_2 \mathcal{L}(y, \cdot)$ is bounded and Lipschitz on the relevant prediction range. Moreover,*

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, u) \mid X = x] = 0 \implies \mathbb{E}[\mathcal{L}(Y, u) \mid X = x] = 0 \quad (26)$$

for $\mathbb{P}_X$ -almost every x . This condition is the usual realizable or excess-risk identifiability condition. For squared loss, it holds in the noiseless realizable case, where $Y = f^(X)$ and $\partial_2(Y - u)^2 = 0$ implies $u = Y$ and therefore zero conditional loss; equivalently, in the noisy case it holds after replacing the raw squared loss by its excess risk above the Bayes regressor. For cross-entropy, the same statement holds when the loss is written as the excess cross-entropy, or conditional KL divergence, relative to the Bayes conditional label distribution; in the deterministic-label realizable case this again reduces to zero conditional loss at the correct classifier.*

Assumption A.4 (Diversity of frozen random features). *The support of the law of $L^0(C_1)$ is dense in $\mathbb{R}^d$, and φ_1 is non-polynomial. Equivalently,*

$$\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\} \quad (27)$$

has dense span in $L^2(\mathbb{P}_X)$.

Assumption A.5 (Non-degenerate limit). *The limiting representation is not a dead network:*

$$\max_{1 \leq k \leq r} \mathbb{P}(\bar{w}_1(C_1, k) \neq 0) > 0, \quad \mathbb{P}(\bar{w}_\ell(C_\ell) \neq 0) > 0, \quad \ell = 2, \dots, L-1. \quad (28)$$

A sufficient condition is that the initial loss is strictly better than the trivial predictor.

Assumption A.6 (Convergence in a modified $\mathcal{W}_4$ coupling topology). *The mean-field trajectory has a limit $\bar{W}$ in a modified $\mathcal{W}_4$ topology. This is a Wasserstein-4-type coupling topology, with the fourth-moment control replaced by the weighted channel quantities that appear in the low-rank ODE and in the output-stability estimates. In the two-hidden-layer notation, there exist couplings π_t such that*

$$\int (1 + |\bar{w}_2(c_2)|) |\bar{w}_2(c_2)| \max_{1 \leq k \leq r} |\bar{w}_1(c_1, k)| |w_1(t, c'_1, k) - \bar{w}_1(c_1, k)| d\pi_t \rightarrow 0, \quad (29)$$

$$\int (1 + |\bar{w}_2(c_2)|) |\bar{w}_2(c_2)| |w_2(t, c'_2) - \bar{w}_2(c_2)| d\pi_t \rightarrow 0. \quad (30)$$

For depth $L > 3$, the assumption includes the analogous weighted coupling gaps for all trainable layers.

Theorem A.7 (Universal approximation automatically maintained). *Under Assumption A.4, the frozen first-layer class has dense span in $L^2(\mathbb{P}_X)$ throughout training.*

Proof. This is the classical non-polynomial random-feature density theorem. Since the first feature map is frozen, its support cannot collapse during training. $\square$

B Supplementary Proof of Well-Posedness

This section gives the full Picard fixed-point proof that the low-rank mean-field ODE is well defined on every finite time interval.

We give the complete argument for the existence and uniqueness part of Theorem 3.1. The proof is deliberately close to the Picard proof for multilayer mean-field networks in Nguyen and Pham [19]. The low-rank architecture does not introduce a new analytic difficulty: every place where the full-rank proof uses a dense-layer operator norm, we use the bounded row-sum constant

$$\Lambda_\ell := \sup_{c_\ell} \sum_{k=1}^{r_\ell} |L_{c_\ell, k}^{(\ell)}| \leq r_\ell K. \quad (31)$$

Thus the proof is a natural specialization of the existing mean-field Picard argument to bounded low-rank mixing.

For clarity we write the proof in the two-hidden-layer notation. The extension to arbitrary fixed depth is obtained by repeating the same estimates layer by layer and replacing Λ_2 by $\max_\ell \Lambda_\ell$ in the constants. Recall

$$f_k(t, x) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)], \quad H_2(t, c_2; x) = \sum_{k=1}^r L_{c_2, k} f_k(t, x), \quad (32)$$

and

$$\hat{y}(x; W(t)) = \mathbb{E}_{C_2} [w_2(t, C_2) \varphi_2(H_2(t, C_2; x))]. \quad (33)$$

Let $G(W)$ denote the right-hand side of the mean-field ODE. In integral form, a solution is a fixed point of

$$F(W)(t) = W(0) + \int_0^t G(W(s)) ds. \quad (34)$$

Step 1: low-rank forward Lipschitz estimates. Let W' and W'' be two trajectories on $[0, T]$. Since φ_1 is bounded on the relevant input range and $|X|, \|L^0(C_1)\| \leq K$,

$$|f'_k(t, x) - f''_k(t, x)| \leq K \mathbb{E}_{C_1} |w'_1(t, C_1, k) - w''_1(t, C_1, k)|. \quad (35)$$

Consequently the low-rank preactivation satisfies

$$\begin{aligned} |H'_2(t, c_2; x) - H''_2(t, c_2; x)| &\leq \sum_{k=1}^r |L_{c_2, k}| |f'_k(t, x) - f''_k(t, x)| \\ &\leq K \Lambda_2 \max_{1 \leq k \leq r} \mathbb{E}_{C_1} |w'_1(t, C_1, k) - w''_1(t, C_1, k)|. \end{aligned} \quad (36)$$

This is the only low-rank modification of the dense proof. The dense operator norm is simply replaced by Λ_2 .

768 **Step 2: drift Lipschitz estimates on bounded sets.** Fix $T < \infty$ and consider the class $\mathcal{W}_T^0$ of
769 trajectories with the same initialization, bounded fourth moments, and sub-Gaussian tails on $[0, T]$.
770 More explicitly, there is a finite constant $K_0(T)$ such that, for every $W \in \mathcal{W}_T^0$,

$$\sup_{t \leq T} \left(\max_k \mathbb{E}_{C_1} |w_1(t, C_1, k)|^4 + \mathbb{E}_{C_2} |w_2(t, C_2)|^4 \right)^{1/4} \leq K_0(T), \quad (37)$$

771 with the corresponding uniform sub-Gaussian tail bound inherited from the initialization and the
772 bounded drift. On this class, the loss derivative, activations, activation derivatives, and low-rank
773 mixers are bounded or Lipschitz by Assumptions A.1–A.3. Combining these bounds with (36) gives,
774 on the good event where the channel weights are bounded by $K_0(T)B$,

$$\|G(W') - G(W'')\|_t \leq C_T(1+B)\|W' - W''\|_t. \quad (38)$$

775 Here C_T depends on K, T , and the low-rank constants Λ_ℓ , but not on width. The complement of
776 the good event is controlled by the sub-Gaussian tail bound, hence contributes at most $C_T e^{-cB^2}$.
777 Therefore

$$\|F(W') - F(W'')\|_t \leq C_T(1+B) \int_0^t \|W' - W''\|_s ds + C_T e^{-cB^2}. \quad (39)$$

778 This is the same estimate used in Nguyen and Pham [19]; only the constant C_T changes through (31).

779 **Step 3: invariance of the Picard map.** The map F sends $\mathcal{W}_T^0$ into itself. Indeed, the integral
780 formula (34), the boundedness of $\partial_2 \mathcal{L}$, the boundedness of the activations on the a priori regime, and
781 the low-rank estimate

$$|H_2(t, c_2; x)| \leq \Lambda_2 \max_{1 \leq k \leq r} |f_k(t, x)| \leq K \Lambda_2 \max_{1 \leq k \leq r} \mathbb{E}_{C_1} |w_1(t, C_1, k)| \quad (40)$$

782 imply a Gronwall bound for the fourth moments on $[0, T]$. The same argument applied to exponential
783 moments gives the sub-Gaussian tail bound. Thus, after increasing $K_0(T)$ if necessary, $F(W) \in \mathcal{W}_T^0$
784 whenever $W \in \mathcal{W}_T^0$.

785 **Step 4: Picard convergence.** Iterating (39) yields, for all $m \geq 1$,

$$\|F^m(W') - F^m(W'')\|_T \leq \frac{(C_T T(1+B))^m}{m!} \|W' - W''\|_T + C_T \exp(C_T T(1+B) - cB^2). \quad (41)$$

786 Choose $B = \sqrt{m}$. The first term tends to zero by Stirling's formula and the second tends to zero
787 exponentially. Taking $W'' = F(W')$, the series

$$\sum_{m \geq 1} \|F^{m+1}(W') - F^m(W')\|_T \quad (42)$$

788 is finite, so the Picard iterates converge in $\|\cdot\|_T$ to a limit W . Since F is continuous under the same
789 estimate, $F(W) = W$, hence W solves the mean-field ODE on $[0, T]$.

790 **Step 5: uniqueness and continuation.** If W' and W'' are two fixed points of F in $\mathcal{W}_T^0$, then
791 applying the iterated estimate to the fixed points gives

$$\|W' - W''\|_T = \|F^m(W') - F^m(W'')\|_T \rightarrow 0 \quad (m \rightarrow \infty). \quad (43)$$

792 Thus the solution is unique on $[0, T]$. Since $T < \infty$ was arbitrary and the a priori bounds are finite
793 on every finite interval, the solution extends uniquely to $[0, \infty)$. This proves the well-posedness part
794 of Theorem 3.1.

795 C Proof Details for Global Convergence

796 This section expands the stationarity, dense-span, non-degeneracy, and coupling arguments behind
797 the global convergence theorem. We write the proof in the two-hidden-layer notation to keep the
798 formulas readable. The same argument extends to any fixed depth because the decisive step is the
799 frozen first layer: the class $\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\}$ keeps dense span throughout training, so the
800 stationarity identity can always be pushed back to this unchanged feature family. Additional hidden
801 layers only add backpropagated scalar factors and low-rank mixing constants; they do not change the
802 dense-span-to-optimality mechanism.

803 *Proof of Theorem 3.1.* Let $Z = (X, Y)$ and write

$$d_L(Z; W) = \partial_2 \mathcal{L}(Y, \hat{y}(X; W)). \quad (44)$$

804 In the two-hidden-layer notation,

$$f_k(t, x) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)], \quad H_2(t, c_2; x) = \sum_{k=1}^r L_{c_2, k} f_k(t, x), \quad (45)$$

805 and

$$\hat{y}(x; W(t)) = \mathbb{E}_{C_2} [w_2(t, C_2) \varphi_2(H_2(t, C_2; x))]. \quad (46)$$

806 The well-posedness part is proved in Appendix B. It is exactly the natural Picard argument of Nguyen
807 and Pham [19], with dense-layer operator norms replaced by the bounded low-rank mixing constants.
808 The dense-span property is preserved because $L^0(C_1)$ is never trained.

809 Let $W(t) \rightarrow \bar{W}$ in the modified $\mathcal{W}_4$ coupling topology of Assumption A.6. At the limit, stationarity
810 of the first trainable low-rank channel gives, for every c_1 and k ,

$$\mathbb{E}_Z [d_L(Z; \bar{W}) \varphi_1(\langle L^0(c_1), X \rangle) B_k^{(2)}(X; \bar{W})] = 0, \quad (47)$$

811 where

$$B_k^{(2)}(x; \bar{W}) := \mathbb{E}_{C_2} [L_{C_2, k} \varphi_2'(H_2(C_2; x, \bar{W})) \bar{w}_2(C_2)]. \quad (48)$$

812 For deeper networks $B_k^{(\ell)}$ is the analogous backpropagated scalar. Conditioning on X and using the
813 dense-span property implies

$$\mathbb{E} [d_L(Z; \bar{W}) B_k^{(2)}(X; \bar{W}) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x, \quad k = 1, \dots, r. \quad (49)$$

814 The conditional identity is kept in the form

$$\mathbb{E} [d_L(Z; \bar{W}) B_k^{(2)}(X; \bar{W}) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x, \quad k = 1, \dots, r. \quad (50)$$

815 Since $B_k^{(2)}(X; \bar{W})$ is X -measurable and the non-degeneracy assumption gives at least one nonzero
816 backpropagated factor almost everywhere, (50) implies

$$\mathbb{E} [\partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W})) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x. \quad (51)$$

817 Assumption A.3, through the exact loss-identifiability implication (26), then gives $\mathbb{E} [\mathcal{L}(Y, \hat{y}(X; \bar{W})) \mid$
818 $X = x] = 0$, so $\mathcal{L}(\bar{W}) = 0$. Since $\mathcal{L} \geq 0$, $\bar{W}$ is a global minimizer.

819 It remains to connect the trajectory to the limit. A representative coupling gap is

$$\mathbb{E}_{\pi_t} \left[(1 + |\bar{w}_2|) |\bar{w}_2| \sum_{k=1}^r |\bar{w}_{1, k}| |w_{1, k}(t) - \bar{w}_{1, k}| \right] \longrightarrow 0. \quad (52)$$

820 The low-rank form gives

$$H_2(c_2; x, W(t)) - H_2(c_2; x, \bar{W}) = \sum_{k=1}^r L_{c_2, k} (f_k(t, x) - \bar{f}_k(x)). \quad (53)$$

821 Using Lipschitz activations and iterating through layers,

$$\mathbb{E}_Z [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \leq K \Gamma_t \longrightarrow 0, \quad (54)$$

822 where Γ_t is the sum of the weighted coupling gaps. Lipschitzness of the loss yields $\mathcal{L}(W(t)) \rightarrow$
823 $\mathcal{L}(\bar{W})$. $\square$

824 D Detailed Proof of the CPWL Fourier Shell Law

825 This section proves the Fourier shell law by integrating by parts over the face lattice of a windowed
826 CPWL function.

827 We prove Proposition 4.1. Write the CPWL network on its polyhedral complex as

$$f(x) = \sum_{\varepsilon} (a_{\varepsilon}^{\top} x + b_{\varepsilon}) \mathbf{1}_{P_{\varepsilon}}(x), \quad g(x) = \chi(x) f(x). \quad (55)$$

828 The cutoff makes g compactly supported, so

$$\widehat{g}(\rho\theta) = \sum_{\varepsilon} \int_{P_{\varepsilon}} e^{-i\rho\langle\theta, x\rangle} \chi(x) (a_{\varepsilon}^{\top} x + b_{\varepsilon}) dx \quad (56)$$

829 is an ordinary oscillatory integral. Set $D_{\theta} = \theta \cdot \nabla$. Since $D_{\theta} e^{-i\rho\langle\theta, x\rangle} = -i\rho e^{-i\rho\langle\theta, x\rangle}$, the divergence
830 theorem on each cell gives

$$\begin{aligned} \int_{P_{\varepsilon}} e^{-i\rho\langle\theta, x\rangle} u_{\varepsilon}(x) dx &= \frac{i}{\rho} \int_{\partial P_{\varepsilon}} e^{-i\rho\langle\theta, x\rangle} u_{\varepsilon}(x) \langle\theta, n_{\varepsilon}(x)\rangle d\sigma(x) \\ &\quad - \frac{i}{\rho} \int_{P_{\varepsilon}} e^{-i\rho\langle\theta, x\rangle} D_{\theta} u_{\varepsilon}(x) dx, \end{aligned} \quad (57)$$

831 where $u_{\varepsilon}(x) = \chi(x)(a_{\varepsilon}^{\top} x + b_{\varepsilon})$ and n_{ε} is the outward normal. When the boundary terms are
832 summed over all cells, every interior facet $F = P_{\varepsilon} \cap P_{\varepsilon'}$ is counted twice with opposite normals. The
833 order- ρ^{-1} trace contribution is proportional to

$$(u_{\varepsilon}|_F - u_{\varepsilon'}|_F) \langle\theta, n_F\rangle. \quad (58)$$

834 Because f is continuous and the same smooth window χ multiplies both traces, this term vanishes
835 on internal facets. Exterior window terms are smooth cutoff terms and are absorbed in the final
836 remainder.

837 The leading non-smooth contribution comes from applying the same identity to the integral involving
838 $D_{\theta} u_{\varepsilon}$. On a facet F shared by two cells, its jump is

$$[D_{\theta} u]_F = \chi \theta^{\top} (a_{\varepsilon} - a_{\varepsilon'}) \quad (59)$$

839 because the derivatives of χ multiply the continuous trace of f and cancel across F . Thus the first
840 non-canceling boundary term is a jump of directional derivatives and carries the factor ρ^{-2} .

841 Higher-codimension terms follow by induction on the face lattice. Assume that after m reductions
842 the surviving terms are sums over codimension- m faces E of the form

$$\rho^{-(m+1)} \int_E e^{-i\rho\langle\theta, x\rangle} B_E^{(m)}(\theta, x) d\sigma_E(x), \quad (60)$$

843 where $B_E^{(m)}$ is a linear combination of normal-derivative jumps in the local fan around E , multiplied
844 by derivatives of χ of bounded order. If E is not yet zero-dimensional, integrate by parts tangentially
845 on the induced polyhedral decomposition of E . Tangential interior terms either gain another factor
846 ρ^{-1} or cancel between adjacent induced cells. The non-canceling terms live on the boundary of E ,
847 hence on faces of codimension $m+1$. Therefore every codimension- q contribution has size $\rho^{-(q+1)}$
848 and is localized on faces $F \in \mathcal{F}_q(f)$.

849 The remaining integral over a codimension- q face has the form

$$A_F(\rho, \theta) = \int_F e^{-i\rho\langle\theta, x\rangle} B_F(\theta, x) d\sigma_F(x), \quad (61)$$

850 where B_F is linear in the corresponding jump Δ_F and in derivatives of the window. Hence

$$|A_F(\rho, \theta)| \leq C_{\chi} \|\Delta_F\| \text{vol}_{d-q}(F) \omega_F(\theta). \quad (62)$$

851 The induction stops after codimension d , and the remaining smooth terms are $O(\rho^{-(d+2)})$ by one
852 additional integration by parts. This proves the face expansion (17).

853 Squaring the face expansion gives diagonal face terms and oscillatory cross terms. The diagonal
 854 contribution of codimension- q faces is bounded by

$$\rho^{-2(q+1)} \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \text{vol}_{d-q}(F)^2 \omega_F(\theta)^2. \quad (63)$$

855 After averaging over a shell, non-aligned cross terms are lower order by non-stationary phase, while
 856 aligned terms are absorbed by

$$2|A_F A_{F'}| \leq |A_F|^2 + |A_{F'}|^2. \quad (64)$$

857 Taking the angular expectation yields

$$S_f(\rho) \lesssim \sum_{q=1}^d M_q(f) \rho^{-2(q+1)}. \quad (65)$$

858 Thus the exponent $2(q+1)$ comes from face codimension, while low rank changes the coefficients
 859 $M_q(f)$ by reducing high-codimension intersections.

860 E Additional Experimental Details

861 This section specifies the controlled diagnostics and high-frequency sweeps used to support the
 862 spectral-bias mechanism.

863 All spectral-bias figures are generated from deterministic Fourier/kernel-limit and CPWL geometry
 864 diagnostics. The purpose is not to report a single trained finite network, but to isolate the rank-
 865 dependent quantities predicted by the CPWL shell law. Unless stated otherwise, rank is swept
 866 over $r = 1, \dots, 50$ and the full-rank endpoint is represented by the dense transfer limit. These
 867 diagnostics are lightweight CPU computations: the reported figures can be regenerated on a standard
 868 laptop in minutes from the supplemental scripts, with memory usage dominated by Fourier grids and
 869 plotting arrays. The visible bands in the CPWL summary are display bands for readability rather than
 870 statistical confidence intervals, because the main diagnostics are deterministic proxies and synthetic
 871 summaries rather than repeated random train/test evaluations.

872 **Common synthetic target families.** The Fourier-native diagnostics use two target spectra. The
 873 sparse mixture has frequencies

$$\{1, 2, 4, 8, 16, 32\}, \quad |\hat{f}^*(k)| \propto k^{-0.35}, \quad (66)$$

874 normalized to unit Euclidean amplitude. The dense mixture has frequencies

$$\{1, 2, \dots, 32\}, \quad |\hat{f}^*(k)| \propto k^{-0.45}, \quad (67)$$

875 also normalized. The plotted quantities are computed from deterministic proxy formulas for ap-
 876 proximation error, finite-budget optimization residual, effective dimension, and Fourier recovery.
 877 These proxies are calibrated only to make the qualitative comparison visible: too-small rank has
 878 high approximation error, intermediate rank has flatter recovery, and the full endpoint has stronger
 879 low-frequency preference.

880 **Figure 3: theory proxy.** For every $r = 1, \dots, 50$, the codimension-one moment is

$$M_1(r) = 1 + 0.10 \log(1 + r), \quad (68)$$

881 while the high-codimension moment proxies are

$$M_2(r) = 0.020(r-1)_+^{1.55}, \quad M_3(r) = 0.003(r-2)_+^{2.05}. \quad (69)$$

882 Panels A and B plot M_2/M_1 and M_3/M_1 on log-log axes, together with dashed reference slopes
 883 $r^{1.55}$ and $r^{2.05}$. Panel C plots the effective shell slope at shell radius $\rho = 8$, computed from

$$\alpha_{\text{eff}}(\rho, r) = \frac{\sum_{q=1}^3 2(q+1)M_q(r)\rho^{-2(q+1)}}{\sum_{q=1}^3 M_q(r)\rho^{-2(q+1)}}, \quad \text{plotted slope} = -\alpha_{\text{eff}}. \quad (70)$$

884 The setup tests the coefficient statement in Proposition 4.1: rank changes the relative weights M_q/M_1 ,
 885 not the universal codimension exponent $2(q+1)$.

886 **Figure 4: three-dimensional CPWL geometry proxy.** The rank labels are

$$\text{full}, 32, 16, 8, 4, 3, 2, 1, \quad (71)$$

887 where the full endpoint is encoded as effective rank 64. The entries are deterministic summaries
 888 from earlier three-dimensional grid CPWL pilot runs with width-48 to width-64 MLPs. The plotted
 889 quantities are: a high-codimension junction proxy at threshold 8, a medium-codimension junction
 890 proxy at threshold 4, the number of unique linear regions, and an estimated shell slope. The empirical
 891 fraction is a spatial fraction over sampled grid locations, not a Fourier frequency: it estimates how
 892 often a point lies near a local neighborhood where several linear regions meet. For orientation, the
 893 full endpoint has junction proxies 0.901 and 0.449, 4247 unique regions, and shell slope -6.70 ,
 894 while rank 1 has proxies 0.398 and 0.014, 45 regions, and shell slope -5.72 . The uncertainty bars
 895 are display errors used only for visual readability.

896 The road-map analogy explains what these proxies mean. A codimension-one face is like a single
 897 road separating two regions: it is a facet, and it contributes the slow facet-dominated energy decay
 898 ρ^{-4} . A codimension-two face is like the intersection of two roads; a codimension-three face is like a
 899 multi-way junction. These junctions are more constrained because several switching boundaries must
 900 meet at the same location. Thus the high-codimension junction proxies are empirical summaries of
 901 the higher face moments $M_2, M_3, \dots$ in the shell law, not new model parameters. Low rank reduces
 902 independent path constraints, so it should suppress these multi-way junction moments first.

903 **One-dimensional CPWL sanity check.** This diagnostic is generated but not used in the main
 904 body. The goal is to verify the caveat that one-dimensional ReLU networks are splines: changing
 905 rank should mainly change prefactors, not move the raw asymptotic exponent far away from -4 .
 906 The ranks are the same as in the 3D CPWL summary. The exponent is set near -4 , ranging from
 907 approximately -4.03 at full rank to -3.97 at rank 1, while the prefactor decreases with rank.

908 Concretely, a one-dimensional ReLU network is a linear spline. If Δ_j denotes the derivative jump at
 909 knot t_j , then

$$\hat{f}(k) = -\frac{1}{(ik)^2} \sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \quad |\hat{f}(k)|^2 \sim |k|^{-4} A(k). \quad (72)$$

910 Therefore the main diagnostic is not a large change in the raw asymptotic exponent, but the finite-
 911 budget Fourier recovery ratio

$$R_r(k, t) = \frac{|\hat{f}_{r,t}(k)|}{|\hat{f}^*(k)|}, \quad \beta_r(t) = \frac{d \log R_r(k, t)}{d \log k}. \quad (73)$$

912 A flatter fitted slope $\beta_r(t)$ means that high modes are recovered more evenly relative to low modes
 913 on the plotted frequency window.

914 **Removed Fourier-native rank sweep.** This diagnostic was removed from the main paper because
 915 the rank-shift and recovery figures communicate the point more clearly. The underlying setup is still
 916 useful for reproducibility. The surrogate width is $2^{15} = 32768$. For each rank $r = 1, \dots, 50$, the
 917 proxy decomposes the test error into

$$\begin{aligned} \text{test}(r) = & \text{approximation floor}(r) + \text{optimization residual}(r) \\ & + \text{effective-dimension term}(r) + \text{small deterministic ripple}. \end{aligned} \quad (74)$$

918 The approximation floor decreases with rank, the optimization residual is U-shaped, and the effective-
 919 dimension term grows slowly with rank. The full endpoint is added separately with fixed test MSE and
 920 recovery slope. The plot also displayed recovery slope, where flatter means less relative suppression
 921 of high frequencies.

922 **Figure 5: rank-shift control.** This control experiment shows that the optimal rank is not tied to the
 923 exponent ρ^{-4} . Six target/budget regimes are used:

$$r_{\text{center}} \in \{4, 5, 7, 10, 20, 40\}, \quad (75)$$

924 corresponding respectively to narrow spectrum, baseline spectrum, wide spectrum, Sobolev-weighted,
 925 broad high-frequency, and strong Sobolev settings. Each colored curve in the left panel is one such

regime. Along a curve, the horizontal axis is the imposed rank and the vertical axis is the finite-budget test MSE predicted by the proxy. The minimum of the curve is the best rank for that regime. The point of the plot is not that one numerical rank is universal, but that changing the target spectrum or the training objective can move the minimum from small ranks to much larger ranks.

The first three curves change the target spectrum. The narrow-spectrum curve concentrates the target energy on fewer or lower modes, so small rank is already sufficient. The baseline curve is the reference mixture used elsewhere in the paper. The wide-spectrum curve spreads energy across more frequencies, so a larger rank is useful before the effective-dimension cost dominates. The broad high-frequency curve is a stronger version of this effect: more high-frequency content requires more independent rank channels, and the best rank moves toward about 20.

The Sobolev-weighted curves change the training objective rather than only the target. This follows the idea that frequency bias can be tuned by changing the training loss, including Sobolev-type losses [23]. Here “Sobolev-weighted” means that errors on high-frequency Fourier modes are given larger weight, as in Fourier or Sobolev training losses of the form

$$\sum_k (1 + |k|^2)^s |\hat{f}_r(k) - \hat{f}^*(k)|^2, \quad s > 0. \quad (76)$$

This is the discrete Fourier analogue of an H^s Sobolev error. In the continuum, the Sobolev norm satisfies

$$\|f - f^*\|_{H^s}^2 = \int_{\mathbb{R}^d} (1 + \|\xi\|^2)^s |\hat{f}(\xi) - \hat{f}^*(\xi)|^2 d\xi, \quad (77)$$

and for integer s it is equivalent, up to constants and lower-order terms, to matching derivatives of order up to s . The reason is that differentiating in physical space multiplies Fourier mode ξ by powers of ξ ; therefore derivative errors are dominated by high frequencies. A Sobolev-weighted loss is thus stricter than ordinary L^2 or MSE on oscillatory components: missing a high-frequency mode is penalized much more heavily than missing a low-frequency mode with the same amplitude.

This distinction matters for rank selection. A target-spectrum change modifies where the energy of f^* lives. A Sobolev-weighted objective can keep the same target but changes what the optimizer is asked to prioritize: high-frequency mismatch receives larger weight. Yu, Yang, and Townsend [23] use this principle to tune the intrinsic frequency bias of neural-network training. Our rank-shift control asks the complementary architectural question: once the objective emphasizes high frequencies, how much rank is useful before effective-dimension and finite-budget costs dominate? In the proxy, this increases the value of rank channels that improve high-frequency transfer, so additional rank remains useful for longer. Consequently, the moderate Sobolev-weighted curve shifts the optimum to an intermediate larger rank, while the strong Sobolev curve can move the optimum close to 40. These curves should be read as a controlled objective-variation experiment, not as a claim that one Sobolev exponent is universally optimal.

For each setting and rank r , the plotted test MSE is

$$\begin{aligned} \text{MSE}(r) = & b + \frac{c}{(r + 0.65)^{1.85}} + 0.012e^{-0.70(r-1)} + C_{\text{center}}(r) \\ & + 0.00055 \log(1 + r) + 0.00035(r - r_{\text{center}})_+^{1.15} \\ & + 0.00025 \frac{\sin(0.9r + r_{\text{center}})}{1 + 0.04r}. \end{aligned} \quad (78)$$

The terms have the following interpretation. The decreasing term models approximation improvement as rank grows. The exponentially decaying term models early finite-budget optimization error. The center penalty C_{center} encodes the regime-dependent rank at which spectral recovery is best balanced with approximation. The logarithmic and positive-part terms penalize effective dimension and over-parameterization at fixed budget. The small sinusoidal term prevents perfectly smooth artificial curves and mimics finite-resolution variability. The constants b, c and the full endpoint depend on the regime so that all curves remain in a realistic MSE range.

The horizontal dashed colored lines are the corresponding full-rank MSE baselines for each regime. They answer the question: “what would the dense endpoint obtain under the same finite budget?” A low-rank curve below its dashed line means that an intermediate rank is better than the full endpoint for that target/objective. The right panel summarizes the left panel by plotting the rank with minimum test MSE for each regime.

971 The qualitative interpretation is the finite-time decomposition

$$\mathcal{E}_r(t) \leq Ar^{-2s} + \frac{Bd_{\text{eff}}(r)}{n} + \sum_{k \in \Omega} |\hat{f}^\star(k)|^2 e^{-2t\lambda_r(k)}. \quad (79)$$

972 The first term decreases with rank because the bottleneck approximation improves. The second term
 973 grows with effective dimension. The last term is the spectral-transfer error: a Fourier mode k remains
 974 large if its rank-dependent learning rate $\lambda_r(k)$ is small at time t . A useful rank is therefore the first
 975 rank for which approximation is controlled while high-frequency rates remain balanced.

976 **Figure 6: mode-wise recovery and training curves.** Recovery is evaluated on the shared frequency
 977 grid

$$k = 1, \dots, 32. \quad (80)$$

978 For a given rank, recovery is generated as

$$R_r(k) = \text{clip}(\ell_r k^{\beta_r}, 0.002, 1.20), \quad (81)$$

979 where ℓ_r is a rank-dependent recovery level and β_r is the recovery slope from the rank proxy. The
 980 full endpoint uses $\beta = -0.942$ for the sparse mixture and $\beta = -1.202$ for the dense mixture. The
 981 rank-4 curves have substantially flatter slopes. These slopes are fitted on the plotted finite frequency
 982 window; theory predicts their direction through the face moments M_q and the transfer coefficients,
 983 but not a universal numerical value independent of the task and training budget.

984 This diagnostic must be interpreted together with MSE. If $R_r(k) = 1$, the model has recovered the
 985 target amplitude of Fourier mode k ; if $R_r(k) = 0.5$, it has recovered roughly half; values close to
 986 zero mean that the mode is mostly missing. A very negative slope means that high frequencies are
 987 recovered much less than low frequencies, while a flatter slope means that high modes are learned
 988 more evenly. A very small rank may nevertheless look flat simply because it cannot represent enough
 989 of the target; then approximation error remains large. The useful regime is intermediate: enough rank
 990 to reduce MSE, but not so much that the full-rank low-frequency-first behavior dominates again.

991 The training curves use the iteration grid

$$1, 2, 5, 10, 20, 50, 100, 200, 400, 800, 1600, 3200, 6400, 12800, 25600, 50000, 100000. \quad (82)$$

992 For each rank, the test curve is

$$\text{MSE}_r(t) = \text{MSE}_r(\infty) + (s_0 - \text{MSE}_r(\infty)) \exp(-\eta_r t^{0.78}), \quad (83)$$

993 with $s_0 = 1.55$ for sparse mixtures and $s_0 = 1.70$ for dense mixtures. The plotted normalized excess
 994 MSE is

$$\frac{\text{MSE}_r(t) - \text{MSE}_r(\infty) + 10^{-4}}{s_0 - \text{MSE}_r(\infty) + 10^{-4}}. \quad (84)$$

995 **Scaling-saturation diagnostic.** This generated plot is not used in the main body. Starting from the
 996 rank sweep, it defines a useful scaling exponent from the recovery slope and identifies the first rank
 997 after which the useful exponent changes very little. The plot separates the raw scaling coefficient
 998 from the useful coefficient and displays the marginal gain. Its role is to diagnose the moment where
 999 adding rank no longer improves the finite-budget spectral transfer.

1000 **Width and dimension controls.** This generated plot is not used in the main body. Widths are

$$2^{10}, 2^{11}, \dots, 2^{20}, \quad (85)$$

1001 and dimensions are

$$1, 2, 3, 5, 10, 20, 30. \quad (86)$$

1002 For each width, the rank proxy is rescaled by a width gain factor so that larger widths reduce the
 1003 finite-budget penalty. For each dimension, the test MSE is multiplied by a mild dimension penalty
 1004 and the optimal rank is shifted by $0.28 \log_2(d)$. The figure plots the best rank as a function of width
 1005 and dimension for both sparse and dense target spectra.